

Looping in Java

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Looping (also called iteration) is a programming concept that allows a block of code to execute repeatedly until a specified condition becomes false. Instead of writing the same statements multiple times, makes. Program shorter, easier to read, and more efficient.

Why do we use loops?

Loops are used when we need to:

- Display numbers repeatedly.
- Process element of an array or Collection.
- Read multiple inputs from the user.
- Perform a repetitive calculations.
- Generate patterns.
- Traverse files or data structures.

In Java three types of loops

Entry control loop

1. while loop

2. for loop

Exit control loop.

1. do { } while loop.

Collection loop [List, Dictionary, HashMap, etc.]

1. for each loop. Arrays
for (variable : collection object)

Syntax of while loop:-

```
while (<condition>) {
```

```
// body of the loop.
```

```
}
```

```
package Loops;  
import java.util.*;
```

```
/**  
 * Write a description of class WhileDemo here.  
 *  
 * @author (your name)  
 * @version (a version number or a date)  
 */  
public class WhileDemo  
{  
    public static void main(String[] args){  
        Scanner sc = new Scanner(System.in);
```

```

int start = 1, n;
System.out.print("\nEnter number of times: ");
n = sc.nextInt();
while(start <= n){
    System.out.print("\n" + start+": Hello!");
    start++;
}
}
}

```

For loop % [Skill: If we know number of iteration, then we use for loop],
 For loop is an advance form of while loop.

Syntax:

for (<initializations> ; <condition> ; <increment/decrement>) {

// loop body

}

e.g.

for (i = 1 ; i <= n ; i++) {

// this loop will execute for 'n' number of times

}

When the value of i exceeds the value of 'n' then for loop terminates.

Diagram illustrating the execution of the for loop:

```

for (i = 1 ; i <= n ; i++) {
    3 -> System.out.print("\n" + i);
}

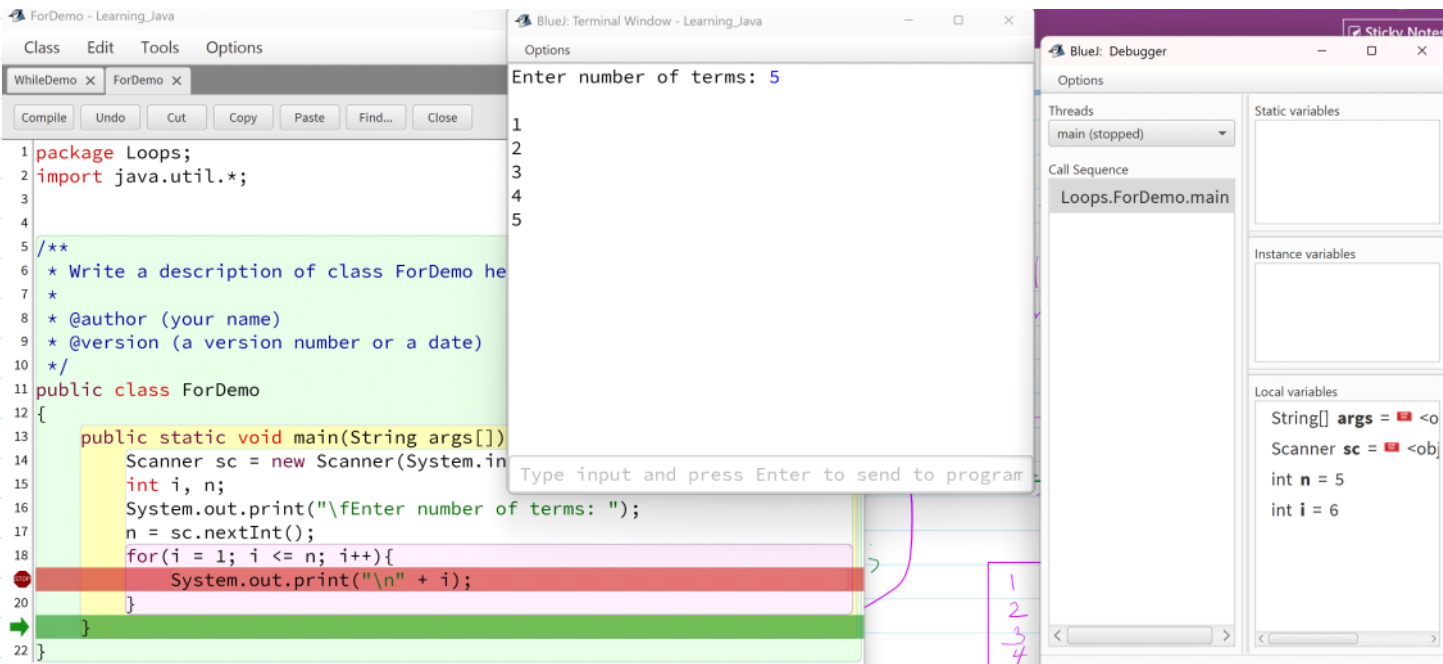
```

The diagram shows the flow of execution with labels F (False) and T (True) indicating the loop condition. The loop body is executed as long as the condition is true.

n = 5

i = 1 2 3 4 5 6

1
2
3
4
5



WAP to input a number and find out how many digits present in it.

Input: Enter any number: 3456

Output: Number of digits: 4

